

EBRAHIM ALIYOU WUDU

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PROFILE

AI researcher focused on building AI systems that remain robust and usable in resource-constrained settings — from ECG diagnostics under real clinical noise to offline, multilingual agricultural tools for East African farmers. Seeking graduate training that bridges technical AI research with human-centered systems design, toward a career developing health and agriculture AI for underserved communities in Ethiopia and across Africa.

EDUCATION

Hindustan Institute of Technology and Science

Chennai, India

Bachelor of Technology in Computer Science and Engineering | 2021 – 2025

- CGPA: 8.79/10 — First Class with Distinction

Relevant coursework: Machine Learning, Deep Learning, Algorithm Design, Data Warehousing and Mining, Big Data Analytics, Theory of Computation, Computer Organization and Architecture, Cloud Deployment

RESEARCH EXPERIENCE

AI Research Intern — University of Galway

May 2026 – Present

sentinelMAS: Evidence-Guided Multi-Agent AI Framework for Histopathology Classification | Supervisor: Dr. Saeed Alsamhi

- Designed an evidence-guided adaptive cascade combining EfficientNet-B0, TinyViT, and the frozen UNI pathology foundation model, with calibrated confidence, conformal uncertainty, and learned modality gating for selective escalation — an architecture built specifically to work under limited training data and compute.
- Achieved 88.77% external accuracy on the patient-disjoint CRC-VAL-HE-7K cohort, versus 78.59% for a standalone CNN and 88.22% for brute-force ensembling, while cutting total cohort inference latency by ~35%.
- Ran component ablations, corruption tests, and multi-method explainability analyses (Grad-CAM++, attention rollout, SHAP) to identify branch contributions and robustness under distribution shift.

AI Research Intern — Haqila Lab, Jimma University

May 2026 – Present

Self-Supervised ECG Representation Learning Under Graded Noise | Advisor: Dr. Gelan Ayana Zewdie

- Benchmarking contrastive, reconstructive, and latent-predictive self-supervised learning methods for ECG representation learning under progressively graded physiological and environmental artifacts, including noise types specific to low-resource clinical settings (e.g., inverter switching noise, 50Hz powerline interference, electrode pop).
- Evaluating robustness, calibration, representation stability, and latent-space behavior to help practitioners choose AI models suited to their deployment setting and safety threshold.
- Findings submitted to the PanAfriCon AI Conference; abstract accepted, full paper under review.

INDEPENDENT RESEARCH

Steering the Machine: Latent State Manipulation in Gemma-3-12B

Aug 2026

Digital Minds Research Sprint, Apart Research

- Investigated whether latent affective steering produces controlled semantic change or broader behavioral degradation, intervening in late-layer residual streams (Layers 32, 40) of Gemma-3-12B.
- Extracted and injected steering vectors at intervention strengths $\alpha \in \{0.0, 0.05, 0.1\}$ with magnitude-matched random-vector controls; evaluated self-reported valence, refusal behavior, sentiment, and GSM8K reasoning under a 256-token generation constraint.
- Found that affective steering produced targeted semantic shifts, while high-magnitude random perturbations caused token looping and structural collapse; refusal-oriented steering frequently produced hyper-rationalization rather than explicit refusal.
- Owned hypothesis design, experiment implementation, representation-engineering code, quantitative analysis, and manuscript preparation.

LEADERSHIP & COMMUNITY IMPACT

Co-Founder & Lead Engineer — GobeZ

May 2026 – Present

Free exam-preparation platform for Ethiopian Grade 12 students | gobez.cloud

- Founded GobeZ after learning that only 3.4% of Ethiopian Grade 12 students passed the national university-entrance exam in 2024, despite a crowded market of existing "AI" study tools — and identified that those tools failed because they surfaced raw performance data instead of actionable, natural-language guidance.

- Led engineering for the platform, including a recommendation engine and Topic Mastery analytics built on 5+ years of solved national exam questions, plus an offline notes feature ("Gobez Bits") for students without reliable connectivity.
- Grew the platform to 640 registered students (500+ monthly active) across 7 partner schools in Addis Ababa; platform is currently offline pending renewed funding.

Team Member — World Technology University Network (WTUN) Hackathon for Climate Action

May 2023

Hosted by University of Bradford

- Joined an international team of four students to design Waste2Wealth, a data-driven circular-economy pipeline connecting organic waste from hotels and supermarkets directly to companies that convert it into new products, targeting the supply and logistics bottleneck in waste processing.
- Placed 2nd of 6 finalist teams from universities worldwide.

Campus Ambassador — GeeksforGeeks

- Conducted technical workshops on AI-powered software engineering practices for 100+ students.

Executive Committee Member — Blue Screen Programming Club

- Organized a national-level coding hackathon (100+ participants) and mentored first-year students in programming.

ENGINEERING EXPERIENCE

Software Development Engineer — Reelo

Apr 2025 – Jan 2026

- Built AWS-backed backend infrastructure processing 100,000+ data points/month; integrated B2B subscription workflows across Chargebee, Stripe, and Razorpay.

Data Engineer Intern — Renan Partners

Feb 2025 – Mar 2025

- Built multi-client ELT pipelines using Meltano, Airbyte, AWS EC2, Shopify extraction, and dbt transformation workflows.

Backend Developer Intern — Reelo

Dec 2023 – May 2024

- Refactored Node.js APIs, reducing response time by 25%; implemented authentication and role-based access control.

Kuza AI — Offline Agricultural AI Assistant

Aug 2026

- Co-founded and engineered an English/Swahili agricultural AI system for East African farmers, addressing a regional farmer-to-agricultural-extension-officer ratio of roughly 1:5000, with a focus on low-resource inference under constrained connectivity, memory, and compute.
- Built multilingual supervised fine-tuning, data-cleaning, model-adaptation, and local-inference pipelines; benchmarked practical deployment trade-offs for edge-oriented AI.
- Selected as a semifinalist in the African Deep Tech Challenge (ADTC 2026) and currently competing to get into top 10 finalists.

AWARDS & HONORS

- Study in India Scholarship, Government of India (2021) — full merit-based government scholarship covering tuition and living costs, awarded amid displacement following the outbreak of war in Ethiopia.
- 2nd Place, WTUN Hackathon for Climate Action (2023) — Waste2Wealth circular-economy prototype.
- 3rd Place, Hack-o-Hits Hackathon (2023) — wearable-assisted medicine intake tracking system.

TECHNICAL SKILLS

Research / ML: PyTorch, TensorFlow, scikit-learn, self-supervised learning, representation learning, interpretability, uncertainty estimation, model evaluation, PEFT/QLoRA, RAG

LLM Research: Representation engineering, steering vectors, LLM evaluation, LangChain, LangSmith

Programming: Python, C++, JavaScript, SQL, GraphQL

Systems: FastAPI, Node.js, AWS, PostgreSQL, MongoDB